STUDENT RESEARCH GROUP ON MEASUREMENT SYSTEMS AND AUTOMATION & RESEARCH ON UNMANNED AERIAL CONTROL SYSTEMS

KN SPA i BLSPS

PARTNERSHIP OPPORTUNITIES

About us

We are a student research group dedicated to advancing the fields of autonomous unmanned systems and artificial intelligence. Our work revolves around integrating UAVs, multi-robot systems, and data analytics with cutting-edge approaches to control and perception.

Operating within a comprehensive project-based framework, we manage the entire lifecycle of our solutions—from initial concept and implementation to rigorous field testing. Our core focus lies in developing tangible, mission-ready systems designed for empirical validation in real-world environments.

Current Research & Projects

MUSC

We are developing a decentralized drone swarm system in which each unit operates autonomously and makes independent decisions based on localized data. By eliminating central control, we have engineered a system where global swarm behavior emerges from peer-to-peer cooperation protocols.

Each drone utilizes its own onboard sensor suite and control system. Localization can be determined through computer vision and triangulation, ensuring the system remains independent of external infrastructure. Our development prioritizes interference resilience and robust performance in contested or GPS-denied environments.

Hawk Eye

We are engineering a stratospheric drone designed to operate at altitudes of 20–25 km. Launched via high-altitude balloon, the system executes autonomous flight upon reaching its target ceiling to conduct environmental research.

Our architecture facilitates a reusable platform, a significant departure from conventional expendable solutions. The mission centers on atmospheric data acquisition and operational reliability under extreme environmental conditions.

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FlySearch

Our team is architecting a framework that integrates Unmanned Aerial Vehicles (UAVs) with Vision-Language Models (VLMs). The drone captures high-fidelity visual and telemetry data, which is then processed by an analytical layer where the model performs semantic scene interpretation to facilitate autonomous decision-making.

The system operates within a continuous perception-action loop. We have implemented a robust data acquisition and analytics pipeline, enabling the development and validation of algorithms in real-world scenarios. Furthermore, we are building a comprehensive infrastructure—encompassing API design, system integration, containerization, and a stable environment for rigorous testing and deployment.

BlueSpark

Our team is engineering a decentralized swarm of Autonomous Underwater Vehicles (AUVs) designed for independent mission execution. Unlike traditional platforms, the system operates without a mothership or continuous human supervision, ensuring full operational autonomy.

The architecture follows a "deploy and forget" model: the user simply defines the mission parameters, after which the system autonomously manages path planning, data acquisition, and inter-unit coordination. Our primary focus is on environmental data collection and subsea mapping, providing scalable solutions for the offshore sector, marine research, and underwater infrastructure inspection.

Snow's Eye

We have designed a ruggedized UAV platform tailored for the world's most demanding environments, including polar regions where GNSS reliability and thermal stability are compromised.

By implementing a multi-layered navigation framework (GNSS, VIO/SLAM), the platform maintains operational safety despite signal interruptions. Our technology has been proven in the field, most notably during expeditions to Svalbard, demonstrating its capability for Arctic environmental monitoring.

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Why partner with us?

Partnering with us offers several strategic advantages, including:

- Support for Emerging Tech Projects: Direct involvement in the growth of high-level, student-led technological initiatives.
- Academic Brand Integration: Establishing a strong presence within the university ecosystem and research community.
- High-Impact Brand Exposure: Visibility across our expanding digital platforms and social media channels.
- Innovation Leadership: Enhancing your corporate image as a key supporter of cutting-edge innovation and R&D.
- Direct Talent Pipeline: Direct access to top-tier engineering students and future industry specialists.

We are committed to scaling our outreach—as our social media presence and promotional activities continue to grow, our partners benefit from increasing engagement, reach, and cross-platform visibility.

Partnership expectation

We are primarily seeking partners for financial or in-kind support. We view our collaborations as mutually beneficial relationships; in exchange for your support, we offer high-level brand exposure and integration across our projects, digital channels, and promotional materials.

Contact details

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